

AGAHI: Adaptive Generation for Academic Hub and Information
(An AI-Powered Study Companion for Pakistani University Students)

Project Proposal

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Abstract

The students studying at universities usually have more study material than they have time to read it all. One course may consist of many weeks of lecture slides, sections from textbooks, notes from instructors, past papers, etc. The task is not only to find the information, but also to determine what to start studying, what subjects need to work on, and whether this material is relevant to the course objectives. Usually, existing study apps perform one task only while general-purpose AI chatbots only answer questions without relying on student's papers or learning experience.

AGAHI is planned to be developed as a web-based learning assistant for university students in Pakistan. The system will accept academic materials in PDF format that include lecture slides, notes, textbooks and past papers. It will help in processing the sources, sorting and summarizing them by using Retrieval Augmented Generation (RAG) technology. The results of the quiz will be evaluated at the topic and sub-topic level so that the system can target difficult subjects and assist in creating the right plan for practicing. The initial revision flow outlines a practice allocation of 70% for weaker subjects and 30% for previously acquired knowledge.

The system also has functions for dealing with common exam preparation issues: research of old exam papers for recurring topics, student weakness heatmap, citation by resources, interactive analytical dashboard, interactive quizzes, and customizable Pomodoro timers tailored for students with ADHD. The goal is to create a unified learning environment where students can transition from learning materials to studying and self-assessing.

1. Introduction

Outcomes of university students' performance, however, often depend less on the lack of information on a particular subject and rather on the abundance of available material. The student might have available information that includes all the lecture slides, hundreds of pages in textbooks, their own notes, and some collections of past exams. When one needs to prepare for the forthcoming examination, all available material falls into one big pool that the student needs to filter through and narrow down.

The challenge becomes even more apparent when it comes to learners with varying levels of understanding of the material. In fact, what might be easy for one student might be extremely difficult for another. Conventional study plans provide students with the same subject matter in the same order. The unfortunate outcome of this is that learners might waste their valuable time on preparing topics they already know while postponing more challenging parts of the syllabus to the end of the preparation process.

The usual AI assistants have made asking questions, summarizing texts, and preparing mock tests much easier. However, such systems might not have knowledge of specific details related to notes, slides, and the previous exams of a certain subject. In other words, it is possible for the systems to provide answers based on non-relevant information, use different terminology, or give explanations that cannot be confirmed based on the previous materials. Therefore, to be suitable for studies, such systems must ensure that the system gives replies based on the uploaded text instead of relying on extra functionalities.

AGAHI solves this problem using the document-based process-oriented educational approach. The educational tool analyzes the resources available for a certain course. The resources are indexed and processed for the pertinent materials to be found at the time of

asking a question. The pertinent information is fed to the language model so that the answer is crafted based on course materials only. Whenever possible, AGAHI provides the answer along with its reference to the original documents and page numbers.

The workflow of research goes beyond just answering questions. The system will summarize uploaded articles and create flashcards and quizzes based on their contents, leaving the content intact and saving the repeated preparation process. After a student takes a quiz, the system will remember the results instead of treating each quiz as an independent event. It will create a comprehensive picture of the student's progress based on the scores and the mistakes on the topic.

The suggested engine for adaptive revision is based on very simple principles in terms of distribution: about 70% of the time will be focused on the 'weak points of the student' while the remaining 30% will cover already known topics for the learner. This percentage is rather a guideline than a fixed percentage suitable for every student. It lets the system organize both remedial learning and material retention within one session.

The analysis of previous examinations is another key element of the proposal. Most students look at past papers manually, trying to understand recurrent questions or clusters of topics. At the same time, the proposed system will analyze all the past submitted assignments together and define the topics frequently represented in the past exams as core components for targeted revision, automating the identification of high-yield subject matter to optimize student preparation. The information gained can be displayed on the dashboard and combined with the student's own record. Thus, an opportunity is provided to distinguish what the course has focused on historically from what the student is struggling with now.

The interface design assumes short and easy interactions instead of one long study page. Sessions connected with quizzes will be based on the idea of bite-sized questions, visible progress, motivation based on streaks, and immediate feedback. These choices are made to make studying easier for students with various study practices and attention spans. A focus timer will support the established intervals and increase the effectiveness of study materials usage without the need to leave the application.

Thus, all these features mean that the program can be described as a course-aware assistant rather than a regular chatbot. The project is aimed at FAST NUCES courses and similar educational settings in Pakistan. It will use English as a main language and text-based formats containing PDF files. As a platform, it will not replace instructors or resources.

2. Goals and Objectives

The goals and objectives of AGAHI are as follows:

1. Create a PDF processing pipeline that can extract and parse content from educational materials, such as lectures and textbooks, into structured topics and subtopics.
2. Construct a context-restricted RAG tool that allows students to access any uploaded document with precise page and source citations in place.
3. Put into operation automated tools to make summaries, flashcards, and summary sheets based on the information extracted from documents.
4. Create an automated quiz producing system that uses distractors from documents with a 70/30 adaptive revision algorithm (70% weak topics, 30% mastered topics).

5. Design a centralized dashboard that will help track the learning progress in terms of Weakness Heatmap or any other relevant graph, chart, or visual representation.
6. Create a way to gather and categorize past examinations according to the topic to find patterns in questions for the exams.
7. Make an interface free from distractions, using immediate visual feedback, streak tracking, and Pomodoro focus timers to support high-engagement study sessions.

3. Scope of the Project

AGAHI is intended for undergraduate and postgraduate students of FAST-NUCES Lahore and other universities in Pakistan. Its objective is to assist the students in organizing their course materials, getting the required concepts, practicing questions and revising the topics where they need maximum focus.

Only text-based PDF files will be accepted by the document processing feature. Scanned PDFs, image-based PDFs which do not have a text layer, handwritten notes and other than PDF files will not be accepted. The upload limit per file is 30 MB or 150 pages whichever is exceeded first. This limit has been set to keep the document processing within reasonable limits of performance on the available cloud infrastructure.

The academic content generated by AGAHI will depend on the documents uploaded by the student. AGAHI will not use any external web searches for generating the answers, summaries, questions or any other study material. AGAHI will support only the English language regarding its user interface and document processing pipeline.

Integration with learning management systems of the academic institutions is excluded from the scope of this program. Also, the system is not expected to perform student enrollment and registration, institution management, payment processes and other associated financial actions. Instead, academic coaching, training, homework assistance and progress tracking are the only areas to be covered by the program.

4. Initial Study and Work Done so Far

We examined existing work tools such as Quizlet, ChatPDF, and Notion AI before finalizing this design to find out their shortcomings in regard to our target use; most of them demand either the production of manually formulated sets of questions [1] or derive answers from general knowledge rather than utilizing the documents of students [2]. AGAHI was designed to bridge this gap. We studied the basic literature on Retrieval Augmented Generation [3] before deciding on a model architecture which would enable the generator to provide answers based on the retrieved context rather than memory loss, which led us to the decision to combine a vector database with the context-only prompting technique [4]. On the practical side, we finalized the technology stack, created a project in Supabase and established an initial database schema, and created the initial version of the PDF ingestion system which is responsible for extracting text and dividing it into chunks that can be embedded [5]. We have also sketched the dashboard and quiz interfaces and collected a small set of sample lecture slides and past papers from our own courses to use for testing the summarization and quiz-generation pipeline as it comes together. Going forward, we expect to have a working web application that a student can realistically use to prepare for a course: uploading their own material, getting summaries and flashcards back within minutes, and working through quizzes that adapt to where they are struggling.

References

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